Spectral Latent Alignment for Cross-Domain Recommendation

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***Abstract*—Cross-domain recommendation aims to improve recommendations in one domain by using knowledge from another related domain. However, many existing approaches either require significant overlap between users or items, or rely on complex deep learning models that are difficult to train and interpret.**

In this work, we propose Spectral Latent Alignment (SLA), a simple and interpretable method for cross-domain recommendation. Instead of directly forcing latent factors or feature distributions to match across domains, our approach focuses on aligning the structural patterns of user–item interaction graphs. We first compute spectral embeddings of each domain using the graph Laplacian, which captures the underlying interaction structure. We then align these embeddings using a small set of anchor correspondences through an orthogonal transformation.

This approach allows knowledge transfer across domains while preserving their individual characteristics. The method is computationally efficient, theoretically grounded, and works effectively even when overlap between domains is minimal.

I. Introduction

Recommender systems are widely used in modern digital platforms such as e-commerce, streaming services, and social media. A common challenge in these systems is **data sparsity**, where the number of user–item interactions is insufficient to learn accurate recommendation models. This issue becomes especially severe in new domains or platforms where only limited interaction data is available.

Cross-domain recommendation (CDR) aims to address this problem by transferring knowledge from a well-established domain (source) to another domain with limited data (target). For example, information from a movie recommendation system may help improve recommendations in a book or music domain.

Traditional approaches for cross-domain recommendation typically rely on shared users or shared items across domains. While effective in some cases, these methods become less useful when the overlap between domains is small. Recent methods based on deep learning attempt to learn domain-invariant representations through neural networks and adversarial training. Although these approaches have shown promising results, they often require large datasets and complex training procedures.

In this work, we explore a simpler perspective. Instead of aligning raw features or forcing latent factors to be identical across domains, we focus on aligning the **interaction structure** of the domains. User–item interactions naturally form graphs whose structural patterns can reveal meaningful relationships between users and items.

Based on this observation, we propose **Spectral Latent Alignment (SLA)**. Our method first computes spectral embeddings of the user–item interaction graph in each domain. These embeddings capture the main structural properties of the interaction network. Using a small set of anchor correspondences between domains, we then align the embedding spaces using a simple orthogonal transformation. This alignment enables effective knowledge transfer while remaining computationally efficient and interpretable.

II. Related Work

Cross-domain recommendation has been widely studied as a way to mitigate data sparsity and cold-start problems in recommender systems. The central idea is to leverage knowledge learned in a data-rich source domain to improve recommendation performance in a target domain with limited interactions [5].

A. Matrix Factorization Approaches

Early work on cross-domain recommendation relied heavily on matrix factorization techniques. One of the most influential approaches is **Collective Matrix Factorization (CMF)**, which jointly factorizes rating matrices from multiple domains while sharing latent representations for overlapping users or items [2]. By enforcing shared latent factors, CMF allows information to propagate across domains. Variants of this approach have been explored in several studies that extend matrix factorization with domain-specific constraints or shared latent spaces [3], [6].

However, these methods rely strongly on the existence of overlapping users or items across domains. When the overlap is small or nonexistent, the shared latent factor assumption becomes difficult to satisfy and transfer performance degrades significantly.

B. Deep Learning Approaches

More recent work has explored deep learning models for cross-domain recommendation. Neural architectures can learn complex nonlinear mappings between latent representations of

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different domains. For example, neural cross-domain recommendation models learn domain-specific embeddings and use neural networks to map these representations across domains [4]. Other approaches integrate autoencoders or multi-task learning frameworks to jointly model multiple domains [10].

While deep neural models can capture complex relationships between domains, they typically require large training datasets and careful hyperparameter tuning. In addition, the resulting models are often difficult to interpret.

C. Adversarial Domain Adaptation

Another line of research applies adversarial learning to reduce domain discrepancy. Domain-Adversarial Neural Networks (DANN) train a feature extractor together with a domain classifier in an adversarial setting so that the learned representations are discriminative for the task while being indistinguishable across domains [1]. This idea has been adopted in several cross-domain recommendation models to learn domain-invariant user representations [7].

Although adversarial methods can reduce domain shift, they often require complex optimization procedures and may suffer from unstable training dynamics. Moreover, aligning feature distributions does not necessarily preserve the structural relationships between users and items that are essential for recommendation tasks.

D. Graph-Based Recommendation Methods

Recently, graph-based approaches have gained significant attention in recommender systems. Since user–item interactions naturally form bipartite graphs, graph neural networks and spectral methods have been used to capture higher-order connectivity patterns in interaction networks [8], [9]. These methods highlight the importance of structural information in recommendation systems.

However, most existing graph-based methods focus on improving recommendation performance within a single domain rather than transferring knowledge across domains.

III. Proposed Method

We propose **Spectral Latent Alignment (SLA)**, a method that transfers knowledge across domains by aligning the structural representations of user–item interaction graphs.

A. Graph Representation

Each domain is represented as a bipartite graph consisting of users and items. Interactions between users and items form the edges of this graph. From this graph we compute the **graph Laplacian**, which captures the connectivity structure of the interaction network.

Using the Laplacian matrix, we compute the top-k eigenvectors to obtain a low-dimensional **spectral embedding** for each node in the graph. These embeddings capture the main structural patterns of the user–item interactions.

B. Anchor Correspondences

To align two domains, we assume the availability of a small set of **anchor correspondences**. These anchors may represent shared users, shared item categories, or other semantic links between domains.

Let U_S and U_T denote the spectral embeddings of the source and target domains respectively. For the anchor nodes, we extract the corresponding embedding rows $U_S^{(a)}$ and $U_T^{(a)}$.

C. Embedding Alignment

We compute an orthogonal transformation matrix Q that aligns the source embeddings with the target embeddings by solving the following optimization problem:

$$\hat{Q} = \arg\min_{Q^T Q = I} \|U_S^{(a)} Q - U_T^{(a)}\|_F^2$$

This is known as the **Orthogonal Procrustes problem** and can be solved efficiently using singular value decomposition (SVD).

The aligned source embeddings are then obtained as

$$\tilde{U}_S = U_S Q$$

After alignment, the source and target embeddings lie in a shared latent space, enabling knowledge transfer between the domains.

D. Recommendation Transfer

Once the embeddings are aligned, a recommendation model trained on the source domain can be applied to the target domain using the aligned representations. This allows information from the source domain to improve recommendations in the target domain.

The main advantages of the proposed approach are:

- It leverages the structural information of user–item interaction graphs.
- It requires only a small number of anchor correspondences.
- It avoids complex adversarial training procedures.
- It is computationally efficient and interpretable.

E. Algorithm

Algorithm 1 Spectral Latent Alignment - Training

Require: Source interaction matrix R_S, target interaction matrix R_T, anchor index set $\mathcal{A}$ (size m), spectral dimension k

Ensure: Trained predictor $\mathcal{M}$, orthogonal alignment $\widehat{Q}$

Construct bipartite graphs G_S and G_T from R_S and R_T
Compute normalized graph Laplacians L_S and L_T
Compute top-k eigenvectors of L_S; form $U_S \in \mathbb{R}^{n_S \times k}$
Compute top-k eigenvectors of L_T; form $U_T \in \mathbb{R}^{n_T \times k}$
Extract anchor row-embeddings:

$$X \leftarrow U_S^{(\mathcal{A})} \in \mathbb{R}^{m \times k}, \quad Y \leftarrow U_T^{(\mathcal{A})} \in \mathbb{R}^{m \times k}$$

Compute matrix $M \leftarrow X^\top Y$
Take SVD $M = U_M \Sigma_M V_M^\top$
Form orthogonal Procrustes estimator $\widehat{Q} \leftarrow U_M V_M^\top$
Form aligned source embeddings $\widetilde{U}_S \leftarrow U_S \widehat{Q}$
Train predictor $\mathcal{M}$ using $\widetilde{U}_S$.
return $\mathcal{M}$ and $\widehat{Q}$ =0

Algorithm 2 Spectral Latent Alignment - Inference

Require: Target interaction matrix R_T, source embedding matrix U_S, orthogonal alignment $\widehat{Q}$ (returned by Algorithm 1), trained predictor $\mathcal{M}$ (returned by Algorithm 1), spectral dimension k, integer K (top-K recommendations)

Ensure: Top-K ranked items for each query user

Construct bipartite graph G_T from R_T
Compute normalized Laplacian L_T of G_T
Compute top-k eigenvectors of L_T; form $U_T \in \mathbb{R}^{n_T \times k}$
for each query user u **do**
  $z_S \leftarrow$ row of U_S corresponding to user u
  $z_u \leftarrow z_S \widehat{Q}$ {map source embedding into target frame}
  $\mathcal{C}_u \leftarrow$ candidate set of target items
  for each item $i \in \mathcal{C}_u$ **do**
    $v_i \leftarrow$ row of U_T corresponding to item i
    $s_{u,i} \leftarrow \mathcal{M}(z_u, v_i)$
  end for
  Sort $\mathcal{C}_u$ by $s_{u,i}$ in descending order
  Output top-K items for user u
end for=0

IV. Theoretical Analysis

In this section we prove a quantitative excess-risk bound for the SLA pipeline under the convention that source embeddings are rotated into the target frame and a predictor is trained on those aligned source embeddings. Concretely, we show that the target risk of the trained model is bounded by the source risk plus two interpretable contributions: (i) a *Procrustes (anchor-estimation) term* that scales as $\sqrt{k}\cdot O\big(\|X^\top E\|_2/\sigma_{\min}(X^\top X)\big)$ and therefore decreases with the number and quality of anchors, and (ii) a *spectral-mismatch term* $\sqrt{k}\,\varepsilon/\gamma$ controlled by the Laplacian perturbation $\varepsilon = \|L_T - L_S\|_2$ and the source spectral gap γ. An additional term Δ_{labels} appears when the conditional label-generating mechanisms differ across domains. This decomposition both justifies SLA's two-step design (spectral embedding followed by orthogonal alignment) and provides actionable diagnostics: the quantities $\varepsilon, \gamma, \sigma_{\min}(X^\top X), \|X^\top E\|_2$ are empirically estimable, so the bound predicts when alignment should help and suggests remedies (e.g., collect more or better anchors, increase anchor diversity, or fine-tune on a small target-labelled set) when transfer is likely to fail.

A. Notation and convention

We adopt the following conventions throughout this proof:

- $L_S, L_T \in \mathbb{R}^{n \times n}$ are the (normalized) Laplacians of the source and target bipartite graphs.
- $U_S, U_T \in \mathbb{R}^{n \times k}$ denote the top-k orthonormal eigenvector matrices (rows are k-dimensional node embeddings).
- Anchors produce $X = U_S^{(a)} \in \mathbb{R}^{m \times k}$ and $Y = U_T^{(a)} \in \mathbb{R}^{m \times k}$.
- $\widehat{Q} \in O(k)$ is the orthogonal Procrustes estimator defined by
$$\widehat{Q} \;=\; \arg\min_{Q \in O(k)} \|XQ - Y\|_F$$
We therefore interpret $\widehat{Q}$ as mapping *source* coordinates into the *target* coordinate frame (i.e. $U_S\widehat{Q}$ is the source embedding expressed in the target frame).
- Let $Q_0 \in O(k)$ denote an orthogonal matrix achieving the minimal alignment between the full eigenspaces:
$$Q_0 \;=\; \arg\min_{Q \in O(k)} \|U_S Q - U_T\|_F$$
- We denote by $\widetilde{U}_S = U_S \widehat{Q}$ the *aligned source* embedding matrix (source $\to$ target).
- f_{tgt} denotes the predictor trained on $\widetilde{U}_S$ (so f_{tgt} expects inputs in the target frame). We also write f_{src} for the predictor trained on raw U_S. They are related by
$$f_{\text{tgt}}(z) = f_{\text{src}}(\widehat{Q}^\top z)$$
- The squared loss is $\ell(f(z), y) = (f(z) - y)^2$ and the domain risk is $R_D(f) = \mathbb{E}_{(z,y)\sim D}[\ell(f(z), y)]$.

B. Assumptions

A1 **Spectral embeddings**: U_S, U_T are the top-k orthonormal eigenvector matrices of L_S and L_T.

A2 **Laplacian perturbation**: $L_T = L_S + \Delta$ with $\|\Delta\|_2 \le \varepsilon$.

A3 **Spectral gap**: The source Laplacian L_S has a spectral gap $\gamma > 0$ at index k (so the top-k eigenspace is isolated).

A4 **Anchors**: There are m anchor correspondences. Let
$$X = U_S^{(a)} \in \mathbb{R}^{m \times k}, \qquad Y = U_T^{(a)} \in \mathbb{R}^{m \times k}$$
be the anchor row-embeddings (rows come from matched nodes). Assume X has full column rank.

A5 **Anchor model**: The anchor residual matrix $E \in \mathbb{R}^{m \times k}$ is defined as:
$$E = Y - XQ_0 = U_T^{(a)} - U_S^{(a)} Q_0$$

A6 **Predictor smoothness & bounded embeddings**: The predictor f_{tgt} is L_f-Lipschitz:

$$|f_{\text{tgt}}(z) - f_{\text{tgt}}(z')| \le L_f \|z - z'\|_2 \quad \forall z, z' \in \mathbb{R}^k$$

and all embeddings satisfy $\|u_{d,i}\|_2 \le B$ for some finite B.

C. Supporting Lemmas

We first state two supporting lemmas that we use in the main proof:

Davis-Kahan Theorem: Under **A1-A3** the principal-angle distance between the top-k eigenspaces satisfies

$$\|\sin\Theta(U_S, U_T)\|_2 \le \frac{\|\Delta\|_2}{\gamma} \le \frac{\varepsilon}{\gamma}$$

Consequently,

$$\min_{Q \in O(k)} \|U_S Q - U_T\|_F \;=\; \|U_S Q_0 - U_T\|_F \;\le\; \sqrt{k}\,\frac{\varepsilon}{\gamma}$$

Remark - This is the standard Davis–Kahan perturbation bound for eigenspaces; the Frobenius inequality follows from $\|\cdot\|_F \le \sqrt{k}\|\cdot\|_2$ for matrices with $rank \le k$.

Procrustes Perturbation: Let $X \in \mathbb{R}^{m\times k}$ have full column rank and let $Y \in \mathbb{R}^{m\times k}$ be anchor rows of the target eigen space. Define Q_0 and E as above so that $Y = XQ_0 - E$. Let $\widehat{Q}$ be the orthogonal Procrustes solution from (X, Y). If

$$\|X^\top E\|2 < \tfrac{1}{2}\,\sigma\min(X^\top X)$$

then

$$\|\widehat{Q} - Q_0\|2 \le \frac{2\|X^\top E\|_2}{\sigma\min(X^\top X)}$$

Sketch - Set $M = X^\top Y = X^\top X Q_0 + X^\top E = A + B$. The Procrustes solution equals the orthogonal (polar) factor $U(M)$ of M, while $Q_0 = U(A)$. Standard perturbation analysis of the polar factor (together with Weyl's inequalities for singular values) yields

$$\|U(A+B) - U(A)\|2 \le \frac{2\|B\|_2}{\sigma\min(A)}$$

when $\|B\|_2 < \frac{1}{2}\sigma\min(A)$. Substitute $A = X^\top X Q_0$ and $B = X^\top E$ and use that $\sigma_{\min}(A) = \sigma_{\min}(X^\top X)$ (right-multiplication by orthogonal Q_0 preserves singular values). This gives the claim.

D. Proof

Let $z_{T,j}$ denote the true target embedding of node j (the j-th row of U_T) and let $\widetilde{z}_{S,j}$ denote the aligned-source embedding for the same index (the j-th row of $\widetilde{U}_S = U_S\widehat{Q}$).The proof proceeds by (i) controlling the change in loss when replacing true target embeddings by aligned-source embeddings, (ii) isolating any label-distribution discrepancy, and (iii) bounding the embedding mismatch via Procrustes and Davis–Kahan estimates.

Step 1 - Relate prediction risk change to embedding error: Since f_{tgt} is L_f-Lipschitz and embeddings are uniformly bounded by B, for every index j and label y we have

$$\big|\ell(f_{tgt}(z_{T,j}), y) - \ell(f_{tgt}(\widetilde{z}_{S,j}), y)\big| \le L_f B \,\|z_{T,j} - \widetilde{z}_{S,j}\|_2$$

Averaging over the target distribution and converting the average of row norms to a Frobenius norm via Cauchy–Schwarz yields the bound

$$\big|R_T^{(\text{true})}(f_{tgt}) - R_T^{(\text{aligned})}(f_{tgt})\big| \le L_f B \frac{1}{\sqrt{n}}\|\widetilde{U}_S - U_T\|_F \tag{1}$$

where

$$R_T^{(\text{true})}(f_{tgt}) = \mathbb{E}_{(z_T,y)\sim T}[\ell(f_{tgt}(z_T), y)]$$

and

$$R_T^{(\text{aligned})}(f_{tgt}) = \mathbb{E}_{(y)\sim T}[\ell(f_{tgt}(\widetilde{z}_S), y)]$$

Step 2 - Bound difference between $R_T^{(\text{true})}(f_{tgt})$ and $R_S(f_{tgt})$: The source risk of f on aligned-source inputs is defined by

$$R_S(f_{tgt}) = \mathbb{E}_{(\widetilde{z}_S,y)\sim S}[\ell(f_{tgt}(\widetilde{z}_S), y)]$$

The difference between the target risk evaluated on aligned inputs and the source risk decomposes into an explicit label-shift term:

$$\begin{aligned}\Delta_{\text{labels}} &:= R_T^{(\text{aligned})}(f_{tgt}) - R_S(f_{tgt}) \\ &= \frac{1}{n}\sum_{j=1}^{n}\Big(\mathbb{E}_{y\sim P_T(\cdot|j)}[\ell(f_{tgt}(\widetilde{z}_{S,j}), y)] \\ &\qquad - \mathbb{E}_{y\sim P_S(\cdot|j)}[\ell(f_{tgt}(\widetilde{z}_{S,j}), y)]\Big)\end{aligned} \tag{2}$$

where $P_T(\cdot \mid j)$ and $P_S(\cdot \mid j)$ are the conditional label distributions at index j in target and source respectively. Combining (1) and (2) gives the exact decomposition

$$R_T^{(\text{true})}(f_{tgt}) - R_S(f_{tgt}) = \big(R_T^{(\text{true})}(f_{tgt}) - R_T^{(\text{aligned})}(f_{tgt})\big) + \Delta_{\text{labels}}$$

$$\Rightarrow\ \big|R_T^{(\text{true})}(f_{tgt}) - R_S(f_{tgt})\big| \le L_f B \frac{1}{\sqrt{n}}\|\widetilde{U}_S - U_T\|_F + |\Delta_{\text{labels}}| \tag{3}$$

If the conditional label mechanism is the same in source and target then $\Delta_{\text{labels}} = 0$ and only the embedding term remains.

Step 3 - Decompose the embedding matrix error into Procrustes error and subspace mismatch: Write $\widetilde{U}_S = U_S\widehat{Q}$ and add-and-subtract $U_S Q_0$ to obtain

$$\begin{aligned}\|\widetilde{U}_S - U_T\|_F &\le \|U_S\widehat{Q} - U_S Q_0\|_F + \|U_S Q_0 - U_T\|_F \\ &\le \sqrt{k}\,\|\widehat{Q} - Q_0\|_2 + \|U_S Q_0 - U_T\|_F\end{aligned} \tag{4}$$

since $\|U_S\|_F = \sqrt{k}$.

Step 4 - Davis-Kahan bound: By the Davis–Kahan inequality,

$$\|U_S Q_0 - U_T\|_F \le \sqrt{k}\,\frac{\varepsilon}{\gamma} \tag{5}$$

Step 5 - Procrustes Perturbation: From the Procrustes/polar perturbation analysis (valid under the mild small-perturbation condition $\|X^\top E\|_2 < \frac{1}{2}\sigma_{\min}(X^\top X)$) we have

$$\|\widehat{Q} - Q_0\|_2 \leq \frac{2\|X^\top E\|_2}{\sigma_{\min}(X^\top X)} \tag{6}$$

Step 6 - Conclusion: Substituting (6) and (5) into (4) yields

$$\|\widetilde{U}_S - U_T\|_F \leq \sqrt{k} \cdot \frac{2\|X^\top E\|_2}{\sigma_{\min}(X^\top X)} + \sqrt{k}\,\frac{\varepsilon}{\gamma}$$

Inserting this bound into (3) and absorbing $1/\sqrt{n}$ into the constant $L_f B$ gives

$$\left|R_T^{(\mathrm{true})}(f_{tgt}) - R_S(f_{tgt})\right| \leq L_f B\Big(\sqrt{k} \cdot \frac{2\|X^\top E\|_2}{\sigma_{\min}(X^\top X)} + \sqrt{k}\frac{\varepsilon}{\gamma}\Big)$$

This completes the proof.

V. Baselines and Hyperparameters

VI. Our Results

VII. Conclusion

References

[1] Yaroslav Ganin, Evgeniya Ustinova, Hana Ajakan, Pascal Germain, Hugo Larochelle, François Laviolette, Mario Marchand, and Victor Lempitsky. Domain-Adversarial Training of Neural Networks. Journal of Machine Learning Research, 2016.

[2] Ajit P. Singh and Geoffrey J. Gordon. Relational Learning via Collective Matrix Factorization. Proceedings of the 14th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD), 2008.

[3] Bin Li, Qiang Yang, and Xiangyang Xue. Can Movies and Books Collaborate? Cross-Domain Collaborative Filtering for Sparsity Reduction. Proceedings of the 21st International Joint Conference on Artificial Intelligence (IJCAI), 2009.

[4] Ali M. Elkahky, Yang Song, and Xiaodong He. A Multi-View Deep Learning Approach for Cross Domain User Modeling in Recommendation Systems. Proceedings of the 24th International World Wide Web Conference (WWW), 2015.

[5] Iván Cantador, Ignacio Fernández-Tobías, Shlomo Berkovsky, and Paolo Cremonesi. Cross-Domain Recommender Systems. In Recommender Systems Handbook, Springer, 2015.

[6] Rong Hu, Liqiang Nie, and Xindong Wu. Personalized Recommendation via Cross-Domain Triadic Factorization. Proceedings of the 22nd International World Wide Web Conference (WWW), 2013.

[7] Fang Liu, Ruiming Tang, Xiuqiang He, Fuzheng Zhang, Ming Zhang, and Shaoping Ma. Cross-Domain Recommendation via Collaborative Deep Learning. Proceedings of the ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD), 2017.

[8] Rex Ying, Ruining He, Kaifeng Chen, Pong Eksombatchai, William L. Hamilton, and Jure Leskovec. Graph Convolutional Neural Networks for Web-Scale Recommender Systems. Proceedings of the 24th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD), 2018.

[9] Xiangnan He, Kuan Deng, Xiang Wang, Yan Li, Yongdong Zhang, and Meng Wang. LightGCN: Simplifying and Powering Graph Convolution Network for Recommendation. Proceedings of the 43rd International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR), 2020.

[10] Zhen Li and others. Deep Cross-Domain Recommendation. ACM Transactions on Information Systems, 2019.